

Threshold Optimality for an MMPP/M/1 Queue with Admission Control

Homework Set 2, Problem 1.3

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May 2026

1 The model

We consider a single-server queue with Markov-modulated Poisson arrivals, exponential service, and admission control. The state is

$$(x, y) \in \mathbb{N}_0 \times \{1, 2\},$$

where x is the number of customers in the system and y is the state of the modulator. In modulator state y , customers arrive at rate λ_y with $\lambda_1 < \lambda_2$. The modulator switches to the other state at rate q , independently of the queue. The server processes at rate μ . An arriving customer can be rejected at cost R per rejection; a holding cost c is paid per unit time per customer in the system. With discount factor $\alpha \in (0, 1)$ and after uniformisation at rate $\gamma = \lambda_2 + \mu + q$, the n -stage value function satisfies, for all $(x, y) \in \mathbb{N}_0 \times \{1, 2\}$,

$$\begin{aligned} V_n(x, y) = & cx + \alpha \Big[\lambda_y \min\{R + V_{n-1}(x, y), V_{n-1}(x+1, y)\} \\ & + \mu V_{n-1}((x-1)^+, y) \\ & + q V_{n-1}(x, y') \\ & + (\lambda_2 - \lambda_y) V_{n-1}(x, y) \Big], \end{aligned} \tag{1}$$

with $V_0 \equiv 0$ and $y' = 3 - y$. The terms inside the bracket correspond, in order, to a (possibly rejected) admission, a potential service completion, a modulator transition, and a dummy self-loop introduced by uniformisation.

The aim of this report is to prove

Theorem 1 (Threshold optimality). *There exist thresholds $U_1, U_2 \in \mathbb{N}_0$ such that, for the discounted cost criterion, it is optimal to reject an arriving customer in state (x, y) if and only if $x \geq U_y$.*

The proof proceeds by induction on n in the event-based dynamic programming framework of Koole (2006); references of the form “Thm. 7. n ” or “Def. 5. n ” below refer to that framework. The strategy is:

1. Decompose (1) into event operators (§2).
2. Identify the functional class to propagate (§3).
3. Verify by induction that each operator preserves that class (§4).
4. Conclude threshold optimality by control monotonicity (§5).

2 Operator decomposition

We identify the value-function components with operators from the framework's catalogue (Defs. 5.1–5.3). The state-space convention treats y as the environment dimension (dimension 0) and x as the single queue dimension (dimension 1).

Discount operator. The direct cost and discount, by Def. 4.1, is

$$T_{\text{disc}}f(x, y) = cx + \alpha f(x, y).$$

Controlled-arrival operator. The admission decision is, by Def. 5.2,

$$T_{\text{CA}(1)}f(x, y) = \min\{R + f(x, y), f(x + 1, y)\},$$

i.e. $T_{\text{CA}(1)}$ with rejection cost R and arrival cost 0.

Single-server departure operator. The service event is, by Def. 5.3,

$$T_{\text{D1}(1)}f(x, y) = f((x - 1)^+, y).$$

Environment operator. The mixing of the four events (admission, departure, modulator-jump, dummy self-loop) and the $y \rightarrow y'$ transition of the modulator are both bundled into the environment operator of Def. 4.2, which in Koole's notation reads

$$T_{\text{env}}(F_1, \dots, F_l)(X) = \sum_{Y \in \mathbb{N}_0} \Lambda(X_0, Y) \sum_{j=1}^l Q^j(X_0, Y) F_j(X^*), \quad (2)$$

where X denotes the full state, X_0 is the environment coordinate, Y is the dummy summation variable ranging over possible new environment values, X^* is X with the 0th coordinate replaced by Y , $\Lambda(X_0, Y)$ is the probability the environment moves from X_0 to Y , and $Q^j(X_0, Y)$ is the conditional probability that mark j (i.e. which of the slots $F_1, \dots, F_l$ fires) occurs given that transition. (Capitals denote Koole's abstract symbols; when we instantiate below for the MMPP, the env coordinate X_0 becomes our $y \in \{1, 2\}$, the dummy summation Y ranges over $\{y, y'\}$, and so on.) In the MMPP instance we have $l = 4$ slots:

- f_1 admission, f_2 departure, f_3 modulator-jump, f_4 dummy self-loop;
- the environment kernel is $\lambda(y, y) = (\lambda_2 + \mu)/\gamma$ (env stays) and $\lambda(y, y') = q/\gamma$ (env jumps), with $\gamma = \lambda_2 + \mu + q$ the uniformisation rate;
- when the env *stays*, the firing mark is admission, departure, or dummy, with conditional probabilities $q^1(y, y) = \lambda_y/(\lambda_2 + \mu)$, $q^2(y, y) = \mu/(\lambda_2 + \mu)$, $q^4(y, y) = (\lambda_2 - \lambda_y)/(\lambda_2 + \mu)$, and $q^3(y, y) = 0$;
- when the env *jumps*, only the modulator-jump mark fires: $q^3(y, y') = 1$, others zero.

Substituting these into (2) and collecting terms gives the expanded form

$$T_{\text{env}}(f_1, f_2, f_3, f_4)(x, y) = \frac{1}{\gamma} \left[\lambda_y f_1(x, y) + \mu f_2(x, y) + q \underbrace{f_3(x, y')}_{\text{via Koole's } X^*} + (\lambda_2 - \lambda_y) f_4(x, y) \right]. \quad (3)$$

The crucial point is that the $y \rightarrow y'$ relabeling of f_3 does *not* require a separate operator: it is produced by the X^* mechanism inside Def. 4.2, which evaluates each slot at the new environment coordinate selected by the outer sum.

Putting it together. Equation (1) can be rewritten in pure Koole form as

$$V_n = T_{\text{disc}}\left(\alpha' T_{\text{env}}[T_{\text{CA}(1)}V_{n-1}, T_{\text{D1}(1)}V_{n-1}, V_{n-1}, V_{n-1}]\right), \quad (4)$$

where $\alpha' = \alpha\gamma$ absorbs the rate factor (the bracket in (1) is exactly γ times the convex combination in (3)). Note that the third V_{n-1} argument appears in operator notation *without any y'* : the $V_{n-1}(x, y')$ term of (1) is recovered by the X^* substitution inside T_{env} when the environment jumps. Concretely, expanding (4) via (3) reproduces (1) term by term.

3 The functional class

By Theorem 3.1 of the framework, if a class $\mathcal{F}$ of functions contains V_0 and is preserved by every operator appearing in (4), then $V_n \in \mathcal{F}$ for all n . We take

$$\mathcal{F} = I(1) \cap CX(1),$$

the class of functions $f : \mathbb{N}_0^2 \rightarrow \mathbb{R}$ that are *non-decreasing in x for each y* (Def. 6.1, $f \in I(1)$) and *convex in x for each y* (Def. 6.3, $f \in CX(1)$):

$$\begin{aligned} f \in I(1) : & f(x, y) \leq f(x+1, y) \quad \forall (x, y) \in \mathbb{N}_0^2, \\ f \in CX(1) : & 2f(x+1, y) \leq f(x, y) + f(x+2, y) \quad \forall (x, y) \in \mathbb{N}_0^2. \end{aligned}$$

The zero function $V_0 \equiv 0$ trivially lies in $\mathcal{F}$.

The choice of $\mathcal{F}$ is motivated by Theorem 8.1 of the framework (stated for $T_{\text{CA}(1)}$ here): if $f \in CX(1)$ then the minimising action in $T_{\text{CA}(1)}f$ is non-increasing in x . Combined with finiteness of the action set, this gives a threshold structure in x for each y , which is exactly the statement of Theorem 1.

4 Propagation lemmas

We show $\mathcal{F} = I(1) \cap CX(1)$ is preserved by every operator in (4).

Lemma 1 (Increasingness propagation). *If $f_1, \dots, f_4 \in I(1)$, then $T_{\text{disc}}f_1$, $T_{\text{CA}(1)}f_1$, $T_{\text{D1}(1)}f_1$, and $T_{\text{env}}(f_1, f_2, f_3, f_4)$ are all in $I(1)$.*

Proof. (a) T_{disc} . We compute

$$T_{\text{disc}}f(x, y) - T_{\text{disc}}f(x-1, y) = c + \alpha [f(x, y) - f(x-1, y)] \geq 0,$$

since $c > 0$, $\alpha > 0$, and $f \in I(1)$. Hence $T_{\text{disc}}f \in I(1)$. (This is the T_{disc} row of Thm. 7.1 with $C(x, y) = cx \in I(1)$.)

(b) $T_{\text{CA}(1)}$. By Thm. 7.2 of the framework, $T_{\text{CA}(1)} : I(1) \rightarrow I(1)$ (the controlled-arrival operator preserves increasingness in the controlled dimension; here, in x).

(c) $T_{\text{D1}(1)}$. By Thm. 7.3, $T_{\text{D1}(1)} : I(1) \rightarrow I(1)$ (the single-server departure operator preserves increasingness).

(d) T_{env} . By Def. 4.2, $T_{\text{env}}(f_1, \dots, f_4)(x, y)$ is a convex combination of the values $f_j(x, y_{\text{new}})$ for $y_{\text{new}} \in \{y, y'\}$ and $j \in \{1, 2, 3, 4\}$, with non-negative weights $\Lambda(y, y_{\text{new}}) Q^j(y, y_{\text{new}})$ (Koole's environment kernel and mark probabilities) that sum to 1. Each f_j is in $I(1)$ by assumption, meaning each y -slice $f_j(\cdot, y_{\text{new}})$ is non-decreasing in x . Closure of $I(1)$ under non-negative convex combinations gives the result. This is the T_{env} row of Thm. 7.1, which is what makes the environment dimension “free” for propagation: as long as the environment kernel Λ does not depend on the queue coordinate x (here it does not — our q is independent of x), T_{env} preserves $I(1)$. $\square$

Lemma 2 (Convexity propagation). *If $f_1, \dots, f_4 \in I(1) \cap CX(1)$, then $T_{\text{disc}}f_1$, $T_{\text{CA}(1)}f_1$, $T_{\text{D1}(1)}f_1$, and $T_{\text{env}}(f_1, f_2, f_3, f_4)$ are all in $I(1) \cap CX(1)$.*

Proof. (a) T_{disc} . Convexity in x :

$$\begin{aligned} T_{\text{disc}}f(x, y) + T_{\text{disc}}f(x+2, y) - 2T_{\text{disc}}f(x+1, y) &= c \cdot [x + (x+2) - 2(x+1)] \\ &\quad + \alpha \cdot [f(x, y) + f(x+2, y) - 2f(x+1, y)] \\ &= 0 + \alpha \cdot (\geq 0) \geq 0, \end{aligned}$$

using $\alpha > 0$ and $f \in CX(1)$ for the second bracket. Hence $T_{\text{disc}}f \in CX(1)$. Combined with Lemma 1(a), $T_{\text{disc}}f \in I(1) \cap CX(1)$.

(b) $T_{\text{CA}(1)}$. By Thm. 7.2, $T_{\text{CA}(1)} : I(1) \cap CX(1) \rightarrow I(1) \cap CX(1)$ (the controlled-arrival operator preserves convexity in the controlled dimension; convexity alone does not propagate through $T_{\text{CA}(1)}$, but the intersection with $I(1)$ does).

(c) $T_{\text{D1}(1)}$. By Thm. 7.3, $T_{\text{D1}(1)} : I(1) \cap CX(1) \rightarrow CX(1)$; combined with Lemma 1(c) we have $T_{\text{D1}(1)}f \in I(1) \cap CX(1)$. (The intersection with $I(1)$ is the side condition for convexity through the single-server departure.)

(d) T_{env} . As in (d) of Lemma 1, the operator is a non-negative convex combination of values $f_j(x, y_{\text{new}})$ for $y_{\text{new}} \in \{y, y'\}$. Each f_j is in $CX(1)$ by assumption, so each slice $f_j(\cdot, y_{\text{new}})$ is convex in x . Closure of $CX(1)$ under non-negative convex combinations gives the result. This is the T_{env} row of Thm. 7.1. $\square$

5 Proof of Theorem 1

Proof of Theorem 1. By induction on n .

Base case. $V_0 \equiv 0 \in I(1) \cap CX(1)$.

Inductive step. Assume $V_{n-1} \in I(1) \cap CX(1)$. By (4),

$$V_n = T_{\text{disc}}\left(\alpha' T_{\text{env}}[T_{\text{CA}(1)}V_{n-1}, T_{\text{D1}(1)}V_{n-1}, V_{n-1}, V_{n-1}]\right).$$

Then:

- By Lemma 2(b), $T_{\text{CA}(1)}V_{n-1} \in I(1) \cap CX(1)$.
- By Lemma 2(c), $T_{\text{D1}(1)}V_{n-1} \in I(1) \cap CX(1)$.
- V_{n-1} itself (the third and fourth slots) is in $I(1) \cap CX(1)$ by the induction hypothesis.
- By Lemma 2(d), T_{env} applied to these four arguments is in $I(1) \cap CX(1)$. Scaling by the positive constant α' preserves the class.
- By Lemma 2(a), T_{disc} applied to it is again in $I(1) \cap CX(1)$.

Hence $V_n \in I(1) \cap CX(1)$.

Threshold structure. For each fixed $y \in \{1, 2\}$, the function $V_n(\cdot, y)$ is convex on $\mathbb{N}_0$. The minimising action in

$$T_{\text{CA}(1)}V_n(x, y) = \min\{R + V_n(x, y), V_n(x+1, y)\}$$

chooses rejection if and only if $V_n(x+1, y) - V_n(x, y) \geq R$. By convexity of $V_n(\cdot, y)$, the increment $V_n(x+1, y) - V_n(x, y)$ is non-decreasing in x , so the set $\{x \in \mathbb{N}_0 : V_n(x+1, y) - V_n(x, y) \geq R\}$ is an upper set: there exists $U_y^{(n)} \in \mathbb{N}_0 \cup \{\infty\}$ such that rejection is optimal at (x, y) if and only if $x \geq U_y^{(n)}$. This is the threshold statement of Thm. 8.1 of the framework, instantiated for $T_{\text{CA}(1)}$ with $f = V_n \in CX(1)$.

Passing to the discounted limit. Under the discount factor $\alpha \in (0, 1)$, the operator $V \mapsto V_n$ is a contraction in the supremum norm on bounded functions, so $V_n \rightarrow V^*$ pointwise

(and uniformly on bounded subsets of $\mathbb{N}_0^2$) as $n \rightarrow \infty$, where V^* is the discounted value function. The class $I(1) \cap CX(1)$ is closed under pointwise limits, hence $V^* \in I(1) \cap CX(1)$. The optimal stationary policy attains the minimum in $T_{CA(1)}V^*$, which by the same convexity argument is threshold in x for each y : there exist $U_1, U_2 \in \mathbb{N}_0$ such that rejection is optimal at (x, y) iff $x \geq U_y$. $\square$

Conclusion. Theorem 1 follows from the event-based decomposition (4) of the value function recursion, the propagation of $I(1) \cap CX(1)$ through each operator (Lemmas 1–2, using the catalogue Thms. 7.1, 7.2, 7.3), and the control-monotonicity result Thm. 8.1 applied to the convex slices $V^*(\cdot, y)$.